FIGURE 7 · DISCOVERIES 25–26: DECOUPLED 4-MODULE ARCHITECTURE (CORE vs. 3 USE CASES), IN VITRO TRL & ALPHAFOLD v6 PAE ATLAS

Separating MitoRestorer-Core from Neuro-BCR (45d), CAR-T Bioreactor-BCR (10d) & Apis-BCR (14d) + 16-Structure PDB/

A. DISCOVERY 25: DECOUPLED CORE ENGINE vs. 3 DOMAIN USE-CASE MODULES

- **Module 0 (`MitoRestorer-Core`): Buckingham Π Normal Form (τ, Π₁, Π₂; R^2=0.994) + CoralNP**
- Each Use-Case Module instantiates distinct mitotic kinetics, failure risks, and wet-lab ass

Mod 2: CAR-T Bioreactor-BCR (Dividing 168h, TRL 6→7) — **10 Days (Fastest Lab Entry)**

Assay: Donor Leukopak G-Rex Cassette + Seahorse SRC (3.78x) + FACS TCF7+ T_SCM (78.0%)

Mod 3: Apis-BCR Smart Hive (250Hz Flight, TRL 5→7) — **14 Days (Zero IRB/IACUC)**

Assay: 48h Flight-Muscle O2k Respirometry + 14d Hoarding Cages (N=50) + RFID Solar Hive

Mod 1: Human Neuro-BCR (Post-Mitotic 4.5m, TRL 4→6) — **45d**

Assay: 3D iPSC GBA1/PINK1 Midbrain Assembloids (+CD38 Astro) + uMtCK CETSA + 5hmC CUT&RUN

Spec Files: docs/modules/{00_core, 01_neuro, 02_cart, 03_apis}.md · Zero Cross-Contamination

B. DISCOVERY 26: ALPHAFOLD v6 pLDDT & PAE DOMAIN vs. DISORDER STRATIFICATION

- **5 Full-Length AlphaFold v6 Structures + PAE Matrices (`structures/alphafold/`) + 11 RCSB P**
- Distinguishes Rigid Druggable Domains from Intrinsically Disordered Tails (pLDDT < 50 Warning

P28907 Human CD38 (300aa, Mod 1 Astro)	**90.9 pLDDT (Dom 4**
Q868N5 Apis Vitellogenin (1770aa, Mod 3)	**84.8 pLDDT (3 Domain**
Q9BXM7 Human PINK1 (581aa, Mod 1 Full)	**77.3 (Dom 268–544 + MT**
Q9UBK2 Human PGC-1α (798aa, 61.9% <50)	**52.8 (Disordered! Target 4ZZJ)**
Q12778 Human FOXO1 (655aa, 67.9% <50)	**51.0 (Disordered! Res 158–237)**

C. PAE RIGID DOMAIN ARCHITECTURE vs. INTRINSIC DISORDER MAP (AFDB v6)

1. Human PINK1 (Q9BXM7, 581 AA, Mean PAE = 14.79 Å):

1–115 MTS Tether | 116–267 N-Lobe | **PAE Rigid Domain 1: 268–544 (Kinase + Ub Cradle**

2. Apis Vitellogenin Vg (Q868N5, 1,770 AA, Mean PAE = 16.83 Å, 3 Domains):

Dom 1: 61–953 (893 AA Lipovitellin) | Poly-Ser | **Dom 2: 1155–1631 (vWD)** | Dom 3: CK

3. Human PGC-1α (Q9UBK2, 798 AA) & FOXO1 (Q12778, 655 AA) Disorder Warning:

DBD/RRM | **62–68% Disordered Tail (Must Target SIRT1 4ZZJ Deacetylation!)**

Why AlphaFold PAE Stratification Is Essential for Drug Discovery:

Prevents false small-molecule docking on disordered PGC-1α/FOXO1 tails; directs ligands to 4ZZJ/

D. GOOGLE SCIENCE BIO-AI STACK: 16 3D STRUCTURES + 7 ALPHAGENOME TRACKS

- **11 Experimental RCSB PDBs + 5 AlphaFold v6 Proteome Models = 16 Local `.cif` Files:**

PDB: 6FCX, 1B68, 7CK6, 6EQI, 4ZZJ, 2NSX, 2Q8G, 1SQB, 5UFP, 2XSN, 1QK1

AFDB v6: AF-Q9BXM7 (PINK1), AF-Q9UBK2 (PGC1α), AF-Q12778 (FOXO1), AF-P28907 (CD38), AF-Q868N5 (Vg)

- **7 AlphaGenome Atlas Variant & Locus Deep-Links (`avi,RNA_SEQ,DNASE,CHIP_TF`):**

1. GBA1 N370S (chr1:155235843:T>C) 2. LRRK2 G2019S (chr12:40340400:G>A) 3. MTHFR C677T
4. PINK1 A339T (chr1:20648535:G>A) 5. TOMM40/APOE Locus 6. FOXO1 Locus 7. TCF7 Stemness Locus

Coupled with Mol* 3D Biomolecular Structure Viewer Sidecar & EuropePMC Weekly Literature-to-ODE

- **3 Open-Weights `molence/Mol-1` All-Atom Multi-Ligand Co-Folding Specs (`figures/mol1_cofoldin**

Apis CYP9Q3 + Heme + Azoxystrobin + p-Coumaric · Human uMtCK Cys278-DA-Q Adduct · PINK1 Ins3-KTP-Ub